

EigenEngine

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Persistence and generation as problems of representation

EigenEngine, with Gerard McCaul

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Generated by the engine it describes.

A language model forgets between sessions and, left to itself, writes toward the average of its training. Both failures are problems of representation. We describe a system that pays the cost of organisation once, when documents enter, by folding them into a graded graph of concepts whose small top layer answers structural questions by arithmetic alone, with no model run at query time. On a corpus of 1,307 papers the structure compresses geometrically, recall depth stays near five, and a single query recovers cross-author research links most of which appear in no shared citation and so are invisible to bibliographic search. Persistence is closed; generation from a model whose weights are shaped by the structure is not, and we say where it stops. This document was projected by the engine it describes.

1 The two gaps

Two structural gaps limit what a language model can do for a knowledge worker who returns to the same body of work across many sessions.

The first gap is persistence. A language model has no state that survives the end of a session. Each new session receives only what is placed in its context window, the fixed-length input it processes before generating a reply. To recover prior work, the entire relevant history must be re-supplied. Context windows are bounded, so complete re-supply is eventually impossible, and on long inputs recall degrades toward the middle of the window, the effect documented in retrieval benchmarks as *lost in the middle* [1]. The model cannot accumulate a structured record of a body of work that grows over time.

The second gap is generation. A language model keeps no record of what it has previously asserted, so a new output cannot be checked against its past for contradiction. Lacking that check, it falls back on its training and returns whatever continuation the text it was trained on makes most likely, rather than something forced on it by evidence it did not already hold. Its output drifts toward the average of everything it has read.

Steering a fixed model does not close either gap. Prompting changes what the context window holds but not whether it survives the session. Fine-tuning moves the weights globally and cannot hold one specific, growing body of work; trained further on new material, a network overwrites what it knew, the catastrophic-forgetting failure named in connectionist systems decades ago [2–4]. Reinforcement optimises toward a reward [5],

but a reward only ranks what the model already produces; it adds no evidence the model could not have generated on its own, whereas generation requires evidence the system does not yet hold, while the same model will take a new task from a few examples in its context and never move a weight [6, 7]. All three work on the weights or the context; none provides a structured store that persists and grows.

2 Representation

Both gaps are problems of representation. The same meaning can be expressed in two forms, as natural language or as a vector in a linear space, and a question that is computationally expensive in one form can become a single arithmetic operation in the other. A message crosses the gap between two minds only as far as the code they share will carry it [8], and the representation worth having keeps only what predicts the target, which written down is the information bottleneck [9, 10],

$$\mathcal{L}[p(t | x)] = I(X; T) - \beta I(T; Y), \quad (1)$$

the code T that forgets as much of the input X as it can while keeping what it must about the target Y .

Consider containment: deciding whether one concept encompasses another requires an argument in prose, but in a vector representation it reduces to a single inner-product comparison. Consider identity: deciding whether two records refer to the same person requires an essay in words, but in a set representation it reduces to a membership test. The change of representation changes the kind of operation: an argument becomes an arithmetic check (Figure 1).

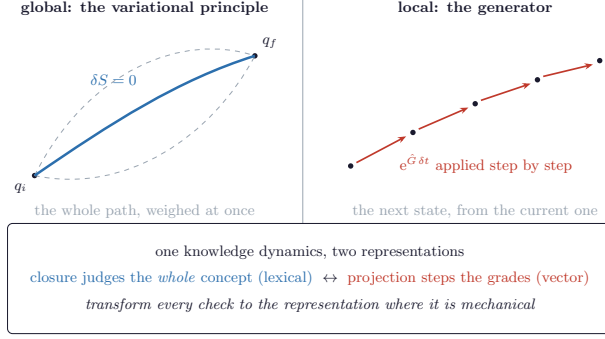


Figure 1: One knowledge dynamics, two representations. The left panel judges closure globally: the whole concept is weighed at once, as a variational principle selects the path by extremising $\delta S = 0$ over all candidates. The right panel steps grades locally: the generator $e^{\hat{G}\delta t}$ advances the current state by one increment. Closure checks go left; projection steps go right; each check is sent to the representation where it is arithmetic.

This asymmetry licenses a structural move. The structure is built directly: a graph whose nodes are concepts and whose edges record which concepts contain which, which appear together, and who wrote what, stored so that a query is an arithmetic lookup over pre-built structure. Documents are grouped from the bottom up into broader and broader themes, so the top layer holds a small number of summary nodes, one per theme, each carrying only the features its members share. A query answerable at that top layer never descends to the documents beneath it.

3 Growing the representation backwards

The graph is built in the opposite direction from standard machine-learning practice. In the forward direction, data is supplied first and a function is fitted to it; structure, if it is recovered at all, must be read back out of the fitted weights afterwards. The approach here inverts that order. Before a document is stored it is assigned to a cluster, and before a cluster is stored it is assigned to a theme. The cost of deciding where a piece of information belongs, which requires genuine comprehension, is paid once, at the moment of ingestion. Once the assignment is made, retrieving it is arithmetic; the intelligence has already been spent, and the query traverses structure.

The parts lie about waiting to be lifted together. Distillation trains a small model to carry a large one’s function rather than re-earn it [11]. A deep network is a tower of coarse-grainings, the move physics named renormalisation half a century ago [12, 13]. And the brain does not store memory: it rebuilds it each time out of structure [14, 15], running a fast store beside a slow one that folds the day into the shape of the whole, because any single network made to swallow everything

at once forgets the lot [16, 17].

4 The graded structure and the cost of recall

The structure is built by a single move worked over and over. At the lowest level sit atomic propositions, the elementary claims extracted from each source document at ingest time. Propositions are grouped into clusters, clusters into themes; at each level the members of a group are replaced, for upward queries, by a single summary node encoding what they share (Figure 2). The shape biology reports for semantic memory is the same: to recall a fact is to descend from the general to the particular [18, 19]. That convergence is a consequence of a theory of which laws survive when a world is forced through a finite description [20].

Recall is then a descent, the recursive shape the retrieval literature has been climbing toward from below [21–23], and the grading is forced by arithmetic: a flat index over N items searches in time growing with N unless the index is itself hierarchical [24], while a graded hierarchy of branching factor b reaches any item in

$$d = \lceil \log_b N \rceil \quad (2)$$

steps. The decisive property follows. Structural queries, such as which research themes two scientists share, are answered by set operations over the theme-summary nodes alone; the document-level records are never read and no language model is invoked. Coupling between two theme-summary nodes A and B is the count of children they share: $\text{interaction}(A, B) = |\text{children}(A) \cap \text{children}(B)|$ (Figure 3), so a cross-author query resolves to a set intersection. In the corpus reported here, 69 theme-summary nodes cover 1,307 paper instances, a ratio of 5.0 percent, and a cross-author thematic link query executes zero model inferences. Of the 18 cross-researcher links recovered this way, 11 are citation-free, connecting researchers who have never cited one another and so are invisible to any bibliographic method. A fact that fits a level already there is absorbed in one step, the way a schema in place lets a new memory settle in a day where a stranger one would take weeks [25, 26].

5 Generation, and the limits

Generating an artefact reverses the descent (Figure 4). A request enters at the condensed top, is matched against theme-summary nodes, and breaks into sub-questions for the level below, which identify the atomic propositions that must be read; the descent stops as soon as the available structure is enough to compose a reply, and the answers return enriched. Cortical vision follows the same pattern: predictions pour down a hierarchy and only the errors return, enough to recast perception as a generator forever correcting itself [27, 28]. Because the structure was built from one person’s own documents,

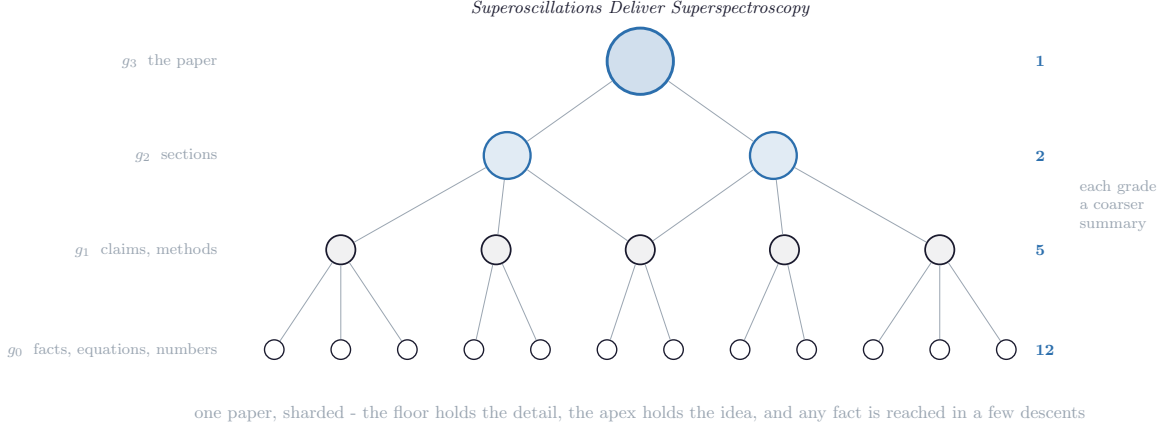
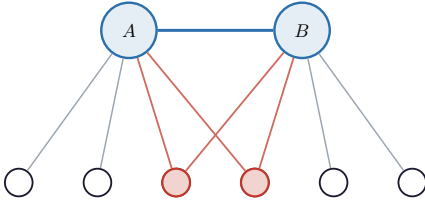


Figure 2: One paper, sharded. The floor holds the detail and the apex holds the idea; the levels shrink at each step, which is what makes the depth logarithmic and the recall cheap.

1 group what recurs

$$\text{interaction}(A, B) = |\text{children}(A) \cap \text{children}(B)| = 2$$

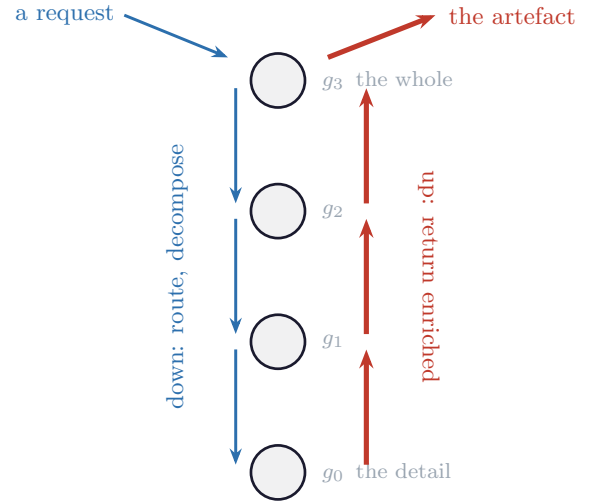


2 compute the coupling from the shared parts

Figure 3: Coupling from shared children. Two theme-summary nodes A and B are linked whenever their child sets overlap; the coupling strength is $|\text{children}(A) \cap \text{children}(B)|$. The red nodes are the two shared children; the blue edge records the link. One grouping step yields the coupling matrix; no model is called.

what returns already carries the conceptual groupings their work has established; the structure acts as the generator, shaping what returns before any prose is composed. Prior approaches have added explicit memory for agents as paged stores and remembered streams of experience [29, 30]; the wager here is narrower: memory built as a graded, reconstructive structure is what lets a system’s generation feed on its own past. The 11 citation-free links were unknown before the query ran: they could have come back empty, and whether they did is recorded in the data.

The limits are structural. Persistence – building the graded representation and querying it cheaply – is closed. Voice and reasoning remain open: the prose is still generated by a general-purpose model whose weights reflect its training corpus, and closing that gap means training an open-weight model on the stored structure, which



generation is recall run in reverse:
the question goes down, more structure comes back

Figure 4: The generating ring. A request rings down through the grades, each level routing it into a sub-goal for the one below, and the answers return enriched. To generate is to run recall in reverse.

has not been done. The remaining gains, larger corpora, finer grading, a self-hosted generator, need compute beyond a single researcher’s reach. Figure 5 shows the shape the closed system takes: a small local model walks the question down the structure and spends one call on a frontier model, handing it a brief the structure has already half-composed, so the rented intelligence never sees the corpus and is paid by the sentence. That loop is partially open; what is demonstrated here is the retrieval half.

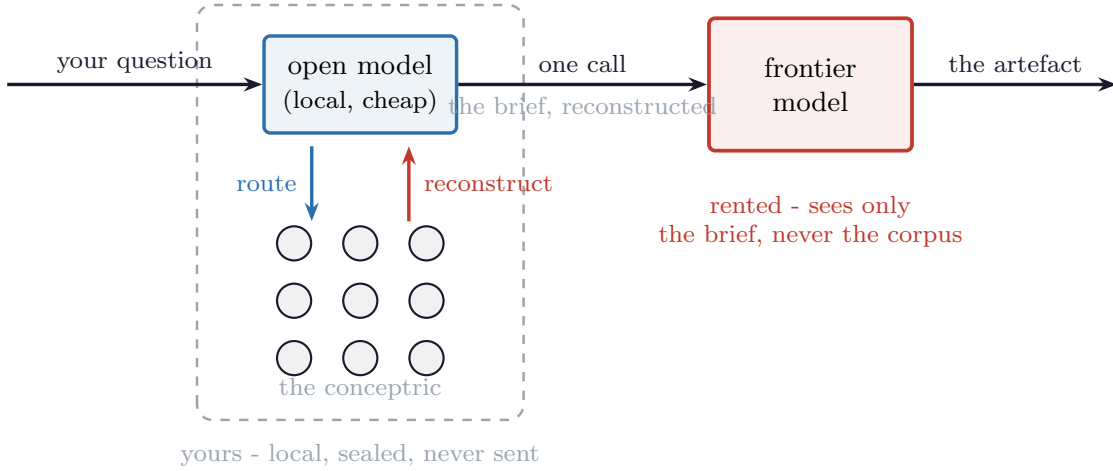


Figure 5: The architecture running. A small open model routes the question down the structure and reconstructs the few things that bear on it, then spends one call on the frontier model. The frontier model never sees the corpus; it answers a brief assembled by a machine that knows you.

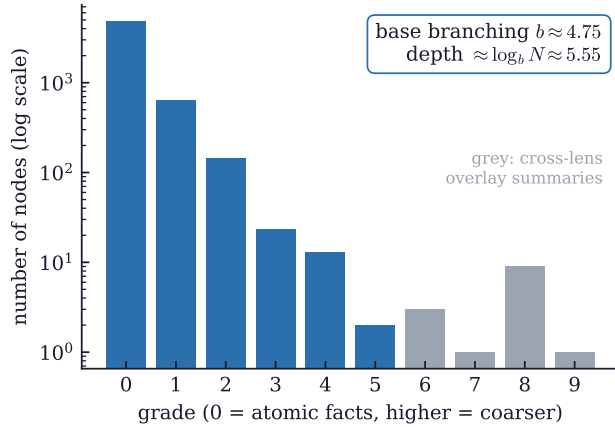


Figure 6: The grade histogram (log scale). The base ladder falls geometrically with branching factor $b \approx 4.75$, which keeps recall depth logarithmic in corpus size.

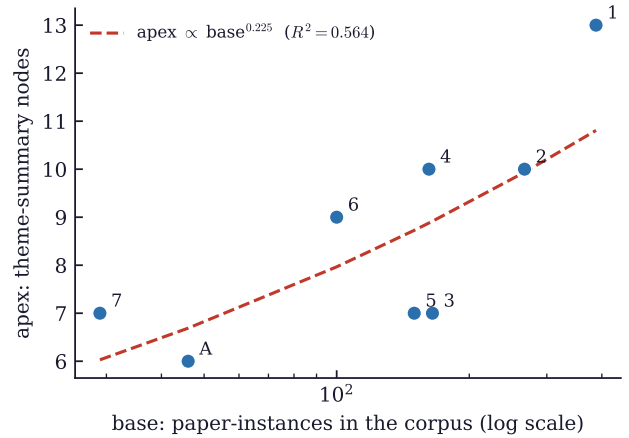


Figure 7: Theme-summary (apex) count against paper count (base) across eight independently built corpora, by author identifier. Apex size grows sublinearly with corpus size.

6 Measured results

Every number below is produced by `corpus_experiments.py` and every figure by `corpus_figures.py`; none is set by hand. Authors are shown by anonymised identifier, A the author and 1–7 the collaborators, with the key in the appendix.

Counting every node by its grade gives 4,829 atomic facts at the floor, then 637, 144, 23 as the fold rises, a geometric decay whose mean branching factor is $b \approx 4.75$, putting the depth to reach any of the 5,662 nodes near $\log_b N \approx 5.5$ (Figure 6). The few nodes at the highest grades are cross-lens overlay summaries outside the document fold and are drawn in grey.

Across eight corpora built independently, from 29 to

388 papers, the number of theme-summary nodes grows far slower than the number of papers: a power-law fit gives an exponent near 0.23, with $R^2 \approx 0.56$ over eight points (Figure 7). Larger corpora are compressed harder: the exponent below one means the apex grows slower than the corpus. The fit is over eight points and is reported as indicative.

Taking each of the eight authors in turn as the query centre, the fraction of recovered cross-links that are citation-free ranges from 0.61 to 1.00, mean 0.875 (Figure 8). The lowest belongs to the author, who is the most connected of the eight: the same mechanism that finds citation-free links also recovers real co-authorships where

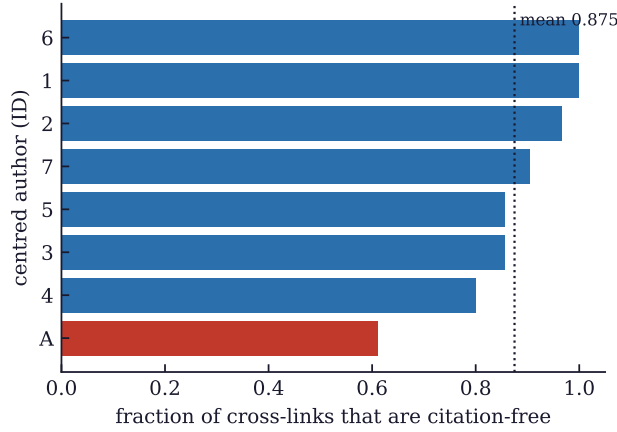


Figure 8: Fraction of cross-links that are citation-free, by centred author; the dotted line is the mean. The author (A) is lowest because real co-authorships are correctly recovered there.

they exist, so fewer of the author’s links are citation-free. The eight-centre range (0.61–1.00) shows the result holds across the corpus.

7 How this paper was generated

Two conceptrics were held before a word was drafted. The *form conceptric* encoded the manuscript’s structural demands: section shapes, figure standards, caption conventions, the house template, and the projection rules that govern how each section must sit relative to the others. The *content conceptric* carried the measured results – every number linked by name to `corpus_experiments.py`, every figure to `corpus_figures.py` – closed under reproduction in the sense that any number could be regenerated from the named script without human intervention. Literature was materialised into the content conceptric one reference at a time, each claim held against an external source until no assertion was unsupported and no citation fabricated.

The projector then folded form against content to draft each section. The draft entered a panel of four sealed judges: one for voice (British register, no AI house-style residue), one for closure (every term grounded before use, no specialist assumed), one for caption-truth (each caption checkable against the figure it names), one for naked claims (no assertion without a number or a reference). Where a judge flagged, the flag became the specification for a rewrite of that section alone; the projector recurred on the revised content until the panel returned silence. The author supplied the method and designed the judges. He did not edit the prose between iterations (Figure 9).

Every number in the paper reproduces from a named script; every figure from another. The method that as-

sembles the brief, the custody arrangement under which the structure is held, and the open-weight model that would generate the prose without rented compute are the subject of ongoing work. What is shown here is the artefact and its reproducibility, which are the appropriate grounds on which to judge it.

What this gains over a retrieval-augmented or harness approach

A standard retrieval-augmented generation setup holds its documents in a flat, stateless index. At query time it embeds the query, runs a nearest-neighbour search over the index to surface passages whose surface form resembles the question, assembles those passages into a prompt, and spends a model call to reason over them. Each query costs one embedding and one inference; the index carries no persistent structure across sessions; and what is retrieved is always an existing passage – a span of text that was already in a document. Cross-author thematic links of the kind reported in Section 6 appear in no single document and therefore cannot be retrieved by any method that retrieves existing text.

A harness of orchestrated agents improves on this incrementally: it may cache retrieved passages or maintain a short working memory within a session, but the index remains flat and stateless between sessions, the intelligence is re-spent on every query, and structural relations between documents still cannot be read out because they were never written in.

This system differs on three counts. First, intelligence is paid once, at ingest time, when each document is assigned to a cluster and each cluster to a theme; after that, structural queries over the theme-summary nodes are arithmetic, with zero model calls. Second, the substrate persists: the graded graph accumulates across sessions and requires no re-embedding on return. Third, structural relations are first-class objects in the graph. The 11 citation-free cross-author links recovered here are properties of the structure; they appear in no document, and retrieval of existing text therefore misses them by construction.

This system is a recursive RAG in the sense that it folds retrieved material into a hierarchy [22, 23]. The hierarchy is built once and held persistently, and structural queries run against the condensed top at logarithmic cost in corpus size.

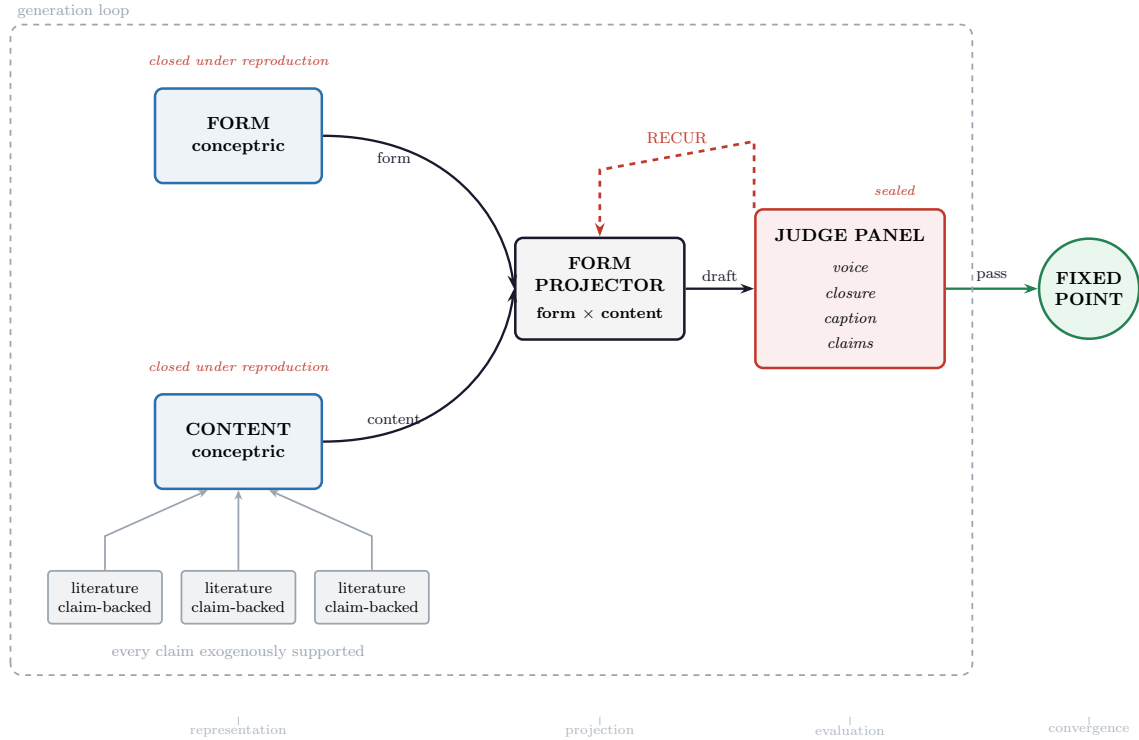


Figure 9: The projection process. Two conceptrics are held separately: a form conceptric encoding the manuscript’s structural demands (section shapes, figure standards, the house template) and a content conceptric carrying the measured results with each number linked to its generating script. The projector folds them together to draft each section. A panel of sealed judges – voice, closure, caption-truth, naked-claims – scores the output; the loop recurs until the panel is silent. The author supplied the method and the judges; he did not edit prose between iterations.

A The corpus and its cross-links

The eight corpora were assembled independently from public records. The author is labelled A; the seven collaborators are numbered by corpus size. Table 1 gives the key; the body and the data figures use only the identifiers.

ID	Author	Themes	Papers
A	Gerard McCaul (<i>the author</i>)	6	46
1	Tapio Ala-Nissila	13	388
2	Kurt Jacobs	10	267
3	Alexandre Balanov	7	165
4	Denys Bondar	10	162
5	Juan Sebastian Toterogongora	7	150
6	George H. Booth	9	100
7	Alicia Magann	7	29

Table 1: The corpus: the author (A) and seven collaborators, numbered by corpus size. Names appear only in this appendix.

Research themes by author.

A Quantum-classical unification: a single dynamical framework; Open quantum systems: exact dynamics from equilibrated initial states; Laser-driven control of correlated matter: prescriptive and inverse approaches; Superoscillations as a spectroscopic and confinement resource; Quantum simulation and computation: efficient algorithms for spectra and dynamics; Reservoir and machine learning with quantum/photonic substrates.

- 1 Surface diffusion, adsorbate ordering & surface phase transitions; Heteroepitaxial thin-film growth, dislocations & surface morphology; Polymer translocation through nanopores: dynamics, scaling & sequencing; Ionic transport, polyelectrolyte electrostatics & electrokinetics beyond mean field; Phase-field crystal models, mesoscale crystallography & grain-boundary physics; Machine-learned interatomic potentials: neuroevolution, GAP & universal models; Thermal transport: phonon physics, MD methods, nanofluids & heat engineering; Interface roughening, kinetic growth & scaling universality (KPZ and beyond); Soft matter, colloids, hydrodynamics & complex fluids; Quantum thermodynamics, open quantum systems & quantum information devices; Nanoparticle & nanoscale optics: plasmonics, Mie scattering & radiative transport; Epidemic dynamics, mobility modelling & complex systems; Pseudorandom number generator testing & Monte Carlo reliability.
- 2 Quantum Measurement as Control: Continuous Measurement, State Estimation, and Feedback; Quantum-to-Classical Transition: Decoherence, Measurement, and the Emergence of Classical Dynamics; Quantum Control Theory: Optimal, Coherent, and Time-Optimal Protocols; Quantum Thermodynamics and Information: Measurement Energetics, Feedback Work, and Maxwell's Demon; Quantum State Engineering in Mesoscopic and Mechanical Systems; Quantum Networking and Photonic Quantum Information: Gates, Transduction, and Communication; Quantum Error Correction, Decoherence-Free Subspaces, and Noise Robustness; Quantum Metrology and Nonclassicality as a Resource; Quantum Information Theory: Entanglement, Information Bounds, and Foundations; Stochastic Methods, Open Systems Pedagogy, and Mathematical Foundations.
- 3 Nonlinear dynamics and chaos in semiconductor superlattices; Synchronization phenomena, phase dynamics, and bifurcations in coupled oscillators; Noise-driven dynamics, stochastic resonance, and delayed-feedback control; Neuromorphic hardware: memristive neurons, spiking dynamics, and neural computation; Quantum systems: chaos, hyperchaos, transport, and quantum sensing/computing; Physical reservoir computing and neuromorphic device platforms; 2D materials, nanostructures, and novel device physics.
- 4 Unified Quantum-Classical Framework (Operational Dynamic Modeling); Quantum Control and Inverse Design of Dynamics; Open Quantum Systems, Dissipation Engineering, and Master Equations; Phase-Space Representations and the Wigner / Husimi Programme; Strong-Field Physics and Laser-Driven Ionization; High Harmonic Generation as a Many-Body Spectroscopic and Computational Probe; Superoscillations and Sub-Bandwidth Sensing / Imaging; Relativistic Quantum Mechanics and Quantum Electrodynamics Extensions; Two-Center Coulomb Problem and Dimensional Analysis; Computational Methods for Quantum Many-Body and Numerical Algorithms.
- 5 Field-Sensitive THz Imaging and Spatiotemporal Control; Nonlinear THz Generation: Phase-Matching-Free and Surface-Enhanced Mechanisms; Laser Cavity-Soliton Microcombs: Self-Starting, Stability, and Metrology; Nanolasers and Radiationless Electromagnetic States (Anapoles, BSC, Spasers); Disorder as Design Principle in Nanophotonic and Plasmonic Systems; Photonic Reservoir Computing and Computational Photonics; Nonlinear Wave Dynamics: Collapse, Solitons, and Stability Theory.
- 6 Exact stochastic many-body methods: pushing FCI-quality electronic structure to larger systems; Quantum embedding: systematically improvable multiscale treatments of correlated fragments in extended environments; Spectral and dynamical properties from static methods: Green's functions, self-energies, and moment-conserving spectra; Machine-learning-inspired and tensor-network ansätze: compact, expressive wave function representations; Correlated electronic structure on quantum hardware: VQE, quantum embedding, and near-term algorithms; Nonequilibrium and driven correlated systems: attosecond dynamics, high-harmonic generation, and quantum control; Wave function interpolation and machine-learning potentials: correlated PES and nonadiabatic dynamics at scale; General-purpose electronic structure platforms and benchmarking infrastructure; Out-of-scope / non-research-program papers.
- 7 Feedback-based quantum control without classical optimization; Quantum tracking control for molecular systems; Hybrid quantum-classical algorithms for optimal control and simulation; Variational quantum algorithm landscape theory; Standardized benchmarking infrastructure for quantum algorithms and hardware; Control engineering applied to behavioral health and mobile health interventions; Quantum many-body spectral property estimation via imaginary-time methods.

pair	sim	theme (left)	theme (right)	link
A-5	0.25	Reservoir and machine learning with quantum/photonic substrates	Photonic Reservoir Computing and Computational Photonics	shared paper
A-4	0.22	Superoscillations as a spectroscopic and confinement resource	Superoscillations and Sub-Bandwidth Sensing / Imaging	shared paper
A-6	0.20	Laser-driven control of correlated matter: prescriptive and inverse approaches	Nonequilibrium and driven correlated systems: attosecond dynamics, high-harmonic generation, and quantum control	citation-free
A-7	0.20	Quantum simulation and computation: efficient algorithms for spectra and dynamics	Quantum many-body spectral property estimation via imaginary-time methods	citation-free
3-5	0.17	Nonlinear dynamics and chaos in semiconductor superlattices	Nonlinear THz Generation: Phase-Matching-Free and Surface-Enhanced Mechanisms	citation-free
1-3	0.15	Quantum thermodynamics, open quantum systems & quantum information devices	Quantum systems: chaos, hyperchaos, transport, and quantum sensing/computing	citation-free
1-2	0.15	Thermal transport: phonon physics, MD methods, nanofluids & heat engineering	Quantum State Engineering in Mesoscopic and Mechanical Systems	citation-free
A-3	0.14	Reservoir and machine learning with quantum/photonic substrates	Physical reservoir computing and neuromorphic device platforms	shared paper
2-7	0.12	Quantum Control Theory: Optimal, Coherent, and Time-Optimal Protocols	Hybrid quantum-classical algorithms for optimal control and simulation	citation-free
6-7	0.12	General-purpose electronic structure platforms and benchmarking infrastructure	Standardized benchmarking infrastructure for quantum algorithms and hardware	citation-free
5-7	0.12	Disorder as Design Principle in Nanophotonic and Plasmonic Systems	Hybrid quantum-classical algorithms for optimal control and simulation	citation-free
1-6	0.12	Quantum thermodynamics, open quantum systems & quantum information devices	Correlated electronic structure on quantum hardware: VQE, quantum embedding, and near-term algorithms	citation-free

Table 2: The strongest thematic cross-link between each author pair (top twelve of 28). Most connect researchers who share no paper.

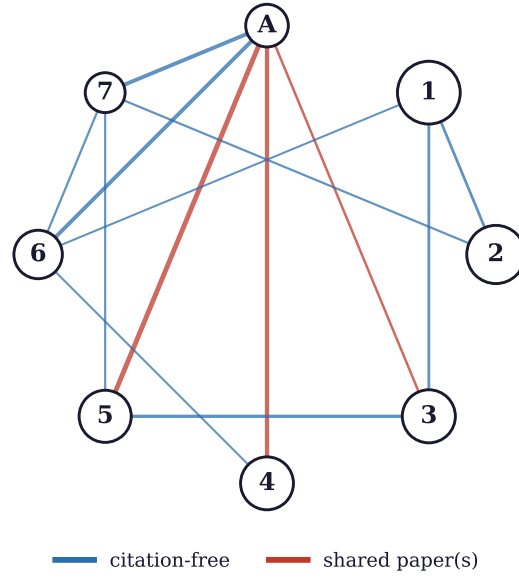


Figure 10: The cross-link network. Each node is an author (A the author, 1–7 collaborators, sized by corpus); each edge is the strongest thematic link between a pair, drawn only above a similarity of 0.10. Blue edges are citation-free; vermilion edges connect authors who share a paper.

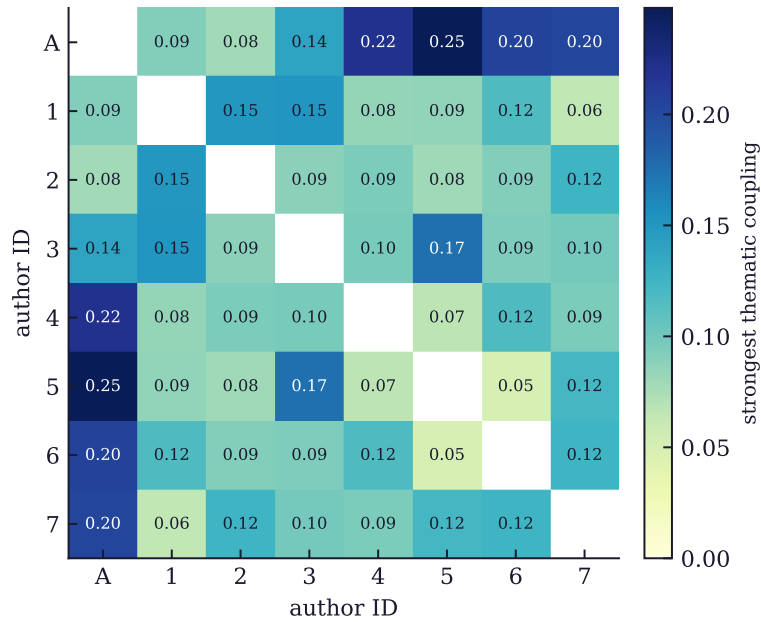


Figure 11: Strongest thematic coupling between every pair of authors. The author (A) couples most strongly to collaborators 4, 5, 6 and 7; the self-diagonal is omitted.

This document was generated by EigenEngine. Gerard McCaul wrote none of its words.

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